# Inequalities — problem 7.35

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Let $x_i > 0$, $x_1 + \cdots + x_n = 1$, and let $s$ be the greatest of the numbers

$$\frac{x_1}{1+x_1}, \ \frac{x_2}{1+x_1+x_2}, \ \frac{x_3}{1+x_1+x_2+x_3}, \ \ldots, \ \frac{x_n}{1+x_1+x_2+\cdots+x_n}.$$

Find the smallest value of $s$. For which $x_1, \ldots, x_n$ will it be assumed ?